Huynh Quoc Viet

AI Researcher & Machine Learning Engineer

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SUMMARY

Artificial intelligence engineering student and research team lead building trustworthy machine learning systems across conformal prediction, multimodal anomaly detection, environmental forecasting, computer vision, and reinforcement learning. Co-author of an accepted AJCAI 2026 poster and AI & Technical Lead of a VAIC 2026 Stage 1 Top-10 team, with hands-on delivery in PyTorch, TensorFlow/Keras, scikit-learn, FastAPI, Streamlit, and reproducible evaluation.

EDUCATION

FPT University Ho Chi Minh City, Vietnam
Bachelor of Engineering in Artificial Intelligence *Expected Jan 2027* | *GPA: 3.12/4.0*

- **Honors:** Honorable Student of the Semester (FALL2025).

HELP University Kuala Lumpur, Malaysia
Certification: English Proficiency Program *Feb 2024*

TECHNICAL SKILLS

Programming: Python, SQL, Java (basic), JavaScript/TypeScript, LaTeX.
Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, Hugging Face, OpenCV, YOLO, RT-DETR.
Methods: Conformal Prediction, Multimodal Anomaly Detection, Computer Vision, Time-Series Forecasting, Reinforcement Learning, Explainable AI.
Data & Engineering: Pandas, NumPy, FastAPI, Streamlit, PostgreSQL, Supabase, Docker, GitHub Actions, Linux, Weights & Biases.

SELECTED PUBLICATIONS

Warning Before the Evening Peak: Health-Centred Air Quality Alerts in New Zealand and Australia
AJCAI 2026 **CORE B** | *Accepted Poster*

- Developed exposure-weighted 1–3 day alert models from 10 years of hourly observations at eight Canterbury stations and four years of Melbourne network data.
- Achieved one-day MAE of 3.19 $\mu g/m^3$ and R^2 of 0.54, improving on persistence by 26%, gradient boosting by 12%, and the strongest sequence baseline by 8%; audited 18 configurations and found exposure protection overestimated by 7.7–18.6 percentage points.

Trustworthy Multimodal Anomaly Detection in Time Series with Conformal Calibration and Counterfactual Attribution
KDD 2027 — Research Track, Cycle 1 **CORE A*** | *Decision pending*

- Built the C-DLE framework and evaluated it across seven real Time-MMD domains; at conformal level 0.10 in a controlled influenza study, fusion raised F1 from 0.641 to 0.757 and TPR from 0.715 to 0.978.
- Added missing-modality calibration, an eight-seed stress test, and a calibration coreset one-tenth the original size; counterfactual attribution reached 94.3% on 122 detected anomalies against injected modality-responsibility ground truth.

Distribution-Free Coverage of Attack Classes on Devices Absent from Training
AISTATS 2027 **CORE A** | *Under Review*

- Evaluated five classifiers and four conformity scores on 68 physical IoT devices, using a device-disjoint 40 train / 14 calibration / 14 test split across eight attack classes.
- Developed a device-quantile calibration rule that kept at least 92.1% of unseen devices within tolerance; established that a 5% target is unattainable with 14 calibration devices and becomes informative from 22 devices.

Stress-Testing Multi-Source Cyanobacterial Bloom Forecasting under Sparse Monitoring
ACIIDS 2027 **CORE B** | *Under Review*

- Built a causal, site-held-out benchmark across seven lakes with an untouched 2023–2024 observed-target holdout ($n = 170$); at seven days, chlorophyll-a history reached weighted F1 0.791 versus 0.688 for full fusion (Holm-adjusted $p = 0.016$).
- Demonstrated that the apparent 14-day fusion benefit was not family-wise significant ($p = 0.096$) and incurred 58.8% alert burden; a provenance audit reduced 22 apparent episodes to three eligible incidents, preventing an unsupported operational-alert claim.

EXPERIENCE

Outlier — Project Aether 06 Jun 2026 – Oct 2026
Contributor — Freelance via Upwork *Vietnamese AI Evaluation*

- Labelled and validated data for Vietnamese-language AI evaluation tasks as a freelance contributor on Project Aether.
- Conducted spoken conversations with two AI models and compared their responses in Vietnamese.
- Evaluated model performance against four rubric dimensions: Audio Quality, Naturalness/Engagement, Utility, and Conversation Dynamics.

SpeedyLabX 05 Nov 2025 – Present
AI Tech Lead — AI Research *Research Team*

- Co-authored conference submissions on health-centred air-quality alerts, conformal multimodal anomaly detection, device-disjoint IoT attack classification, and sparse-monitoring bloom forecasting.
- Reviewed related papers and designed leakage-resistant evaluation protocols, calibration audits, and reproducibility checks to support experiments and manuscript preparation.
- Developed GARCH volatility features, HMM regime models, and directed transition networks for exploratory financial-regime research; the study is currently on hold.

Ky Nguyen So 05 Jan 2026 – 10 May 2026
AI Research Intern — On-the-Job Training (OJT) *FPT University Program*
General Era Digital Solution Joint Stock Company

- Studied Federated Learning (FL), Continual Learning (CL), and Federated Continual Learning (FCL), alongside Edge Computing and Edge AI, for learning from distributed, evolving data.
- Compared federated class-, domain-, and task-continual learning settings through paper reviews, examining data heterogeneity, catastrophic forgetting, and edge resource constraints.
- Documented FCL application studies in intelligent transportation, healthcare, IoT, and UAV networks; maintained research notes and paper summaries throughout the internship.

PROJECTS

FitSync — Personal Training Workspace EXE201 Startup Project
Project Lead — Product & OCR *Sep 2026 – Present*

- Lead a five-member team building an authenticated Next.js and Supabase/PostgreSQL coaching workspace with tenant-isolated profiles and row-level security.
- Delivered single-use trainee invitations, five-field InBody review, private meal-photo check-ins, nutrition drafts, progress history, and an inactivity follow-up queue.
- Own an in-progress OCR workflow with private evidence storage, mandatory coach review, and manual fallback while benchmark design is being finalized.

Adaptive Preference Allocation for Multi-Objective RL REL301m Course Project
Research Engineer *Sep 2026 – Present*

- Implemented a preference-conditioned vector Q-network with per-head Double-DQN targets and adaptive allocation driven by performance gap, disagreement, coverage/recency, and learning-progress signals.
- Designed paired-seed protocols for MO-LunarLander (four objectives) and Minecart (three objectives), each using 32 training, 32 validation, and 100 test preferences with uniform and compute-matched random controls.
- Added replay provenance, budget checks, and reproducible experiment orchestration validated by 21 passing tests; multi-seed scientific experiments remain in progress.

Personal Protective Equipment Detection DPL302m Course Project
Deep Learning Engineer *Sep 2024 – Dec 2024*

- Harmonized nine public datasets into a nine-class detection schema for workers, protective equipment, and missing-equipment violations.
- Benchmarked YOLOv8s/m/l and RT-DETR model families and implemented webcam/video inference with Safe/Unsafe compliance logic.
- Delivered a Streamlit interface supporting image, webcam, and video workflows with annotated-video export.

Face Recognition Attendance System CPV301 Course Project
Model Engineer *Jun 2024 – Aug 2024*

- Built a nine-stage OpenCV/Tkinter pipeline that captures 20 images per person and preprocesses faces at 150×150 using histogram equalization, denoising, and sharpening.
- Combined Haar Cascade detection with LBPH recognition and required two seconds of continuous identity stability before recording attendance.
- Implemented multithreaded UI processing and CSV output with one attendance record per person per day.

COMPETITIONS

NASA Space Apps Challenge 2026 Registered — Upcoming
AI & Tech Lead *Nov 14–15, 2026*

- Lead technical preparation and challenge selection for a five-member multidisciplinary team spanning AI, backend, hardware/IoT, and environmental science/GIS; project outcomes are pending the event.

Vietnam AI Innovation Challenge 2026 — BioListen VN Top 10 — Stage 1
AI & Technical Lead — NeuraX.ai *Jul 2026*

- Led a three-person team through a 48-hour hackathon and reached the Stage 1 Top 10.
- Built a 4,741-sample audio pipeline (3,792 train / 949 validation) and a dual-head EfficientNet-V2-S for three ecological groups and nine threat/background classes using masked multi-task loss.
- Added ONNX export and Grad-CAM evidence for deployment and model inspection.

NASA Space Apps Challenge 2025 — CosmicOptic Ho Chi Minh City
Data & Model Engineer *2025*

- Built a Lightkurve crawler for Kepler/K2 light-curve acquisition, target organization, stitching, and normalization during a 48-hour hackathon.
- Engineered transit/orbital features and integrated FastAPI visualization for CosmicOptic.

FPT Hackathon FPT University
Competitive Programming Participant *May 2025*

- Applied dynamic programming and graph algorithms in an ICPC-style contest under strict time limits.

CERTIFICATIONS

- **IBM:** AI Foundations for Everyone; Data Science Fundamentals with Python and SQL; Full Stack Software Developer.
- **NVIDIA:** Building RAG Agents with LLMs; Developing an AI Background Generator with NIM.